

Extracting The Group Index of Waveguides Built In a Silicon and Silicon Dioxide System

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Abstract—Mach Zehnder Interferometers (MZI) will be designed with the intent of extracting the group index of a variety of waveguide designs based on silicon waveguides and a silicon dioxide cladding surrounding the waveguides. The distance in maxima and minima of the optical power transmitted through the MZI as a function of wavelength will be used to extract the group index of the respective designs. Variations of MZI path length difference to be explored are described below. We find that longer path length differences fell within the expected range of group indices more often and had more repeatable results than short path length differences.

Index Terms—Group Index, MZI, Photonics

I. Introduction

CONTRAST between the index of refraction in a material system enables control over the path that light takes through the system. Silicon and silicon dioxide have enough contrast in their respective indices of refraction, 3.47 and 1.44 respectively, that they can be used in tandem to effectively confine light to propagate along a desired path. With a waveguide designed in silicon and surrounded by a cladding of silicon dioxide, light travelling down the waveguide can be primarily confined to the silicon waveguide. However, the light is not completely confined to the silicon, and some portion of the light's mode extends into silicon dioxide cladding. As a result, the light travelling in the waveguide experiences an effective index of refraction that is somewhere in between the index of refraction of silicon and that of silicon dioxide. How close that effective index is to silicon's or silicon dioxide's bulk index of refraction is a function of geometry, wavelength and mode of the propagating light.

The effective index of light describes how an ideal monochromatic electromagnetic wave, described by a single wavelength, travels through a system. In practice a light pulse will have a distribution of wavelengths centered around a given wavelength and the effective index of

refraction varies as a function wavelength. To describe how a distribution of wavelengths propagates through the system you need to use the group index which accounts for how the effective index varies as a function of wavelength.

$$n_g = n_{eff} - \lambda \frac{dn_{eff}}{d\lambda}$$

Where $\frac{dn_{eff}}{d\lambda}$ will range between the values for bulk silicon and bulk silicon dioxide depending on the geometry of the system as well as the mode of propagating light.

	Bulk Silicon	Bulk Silicon Dioxide
n_{eff}	3.47	1.44
Wavelength Dependence $\frac{dn_{eff}}{d\lambda}$	-7.6e-5 /nm	-1.2e-5 /nm
Temperature Dependence $\frac{dn_{eff}}{dT}$	1.87e-4 /K	8.5e-6 /K

Note that the values for $\frac{dn_{eff}}{d\lambda}$ are negative so the group index is higher than the effective index in this system reflecting the fact that the group velocity is lower than the phase velocity of light.

A technique to measure the group index of a waveguide is to measure the optical power output of an imbalanced Mach-Zehnder interferometer (MZI). An MZI consists of a waveguide in series with a splitter that divides the light into two different waveguide arms. The light propagating along these arms accumulates phase and the quantity of phase accumulated is a function of the lengths of the respective arms, the effective index of the respective arms and the wavelength of light propagating through the arms. The waveguide arms are recombined at a combiner where the waveguides are joined together and the light interferes. The resulting optical power coming out of the combiner will depend on how the light interferes which in turn depends on how much phase the light accumulated on each respective arm. This optical intensity can be described by the equation

$$\frac{I_o}{I_i} = \frac{1}{2} [1 + \cos(\Delta\phi)]$$

Where $\Delta\phi$ is the difference in phase accumulated on each respective arm and can be described by the equation

$$\Delta\phi = \frac{2\pi}{\lambda} (n_{eff,1}L_1 - n_{eff,2}L_2)$$

Or if the effective indices of the two arms are the same

$$\Delta\phi = \frac{2\pi}{\lambda} n_{eff} \Delta L = \beta(\lambda) \Delta L$$

The spacing between minima (or maxima) in equation 1 will occur when $\Delta\phi$ equals integer m values of 2π

$$\Delta\phi(\lambda_m) = 2\pi m$$

Or

$$(\beta_m - \beta_{m+1})\Delta L = 2\pi$$

Assuming

$$\Delta\beta = \frac{2\pi}{\Delta L} \sim -\frac{dB}{d\lambda} \Delta\lambda$$

We can solve for the difference in wavelength between respective minima (or maxima) which is defined as the free spectral range, or FSR.

$$\Delta\lambda \sim -\Delta\beta \left(\frac{dB}{d\lambda}\right)^{-1} = -\frac{2\pi}{\Delta L} \left(\frac{dB}{d\lambda}\right)^{-1}$$

Recognizing

$$\frac{dB}{d\lambda} = \frac{2\pi}{\lambda} \frac{dn}{d\lambda} + 2\pi n(-\lambda^2) = -\frac{2\pi}{\lambda^2} \left(n - \frac{dn}{d\lambda} \lambda\right)$$

Such that the FSR can be defined

$$FSR = \Delta\lambda = \frac{\lambda^2}{\Delta L \left(n - \lambda \frac{dn}{d\lambda}\right)} = \frac{\lambda^2}{\Delta L n_g}$$

Thus, by measuring the FSR of an MZI that has a difference between the length of its arms, we can extract the group index of the waveguide.

$$n_g = \frac{\lambda^2}{\Delta L * FSR}$$

Experimental Design

To design structures capable of extracting the group index we need to select geometries for our MZI's that achieve FSR oscillations in output power that are measurable within the constraints of an experimental system. The optical network analyzer available can produce wavelengths in the region of 1500nm with a bandwidth of 50nm and a typical resolution of 10pm (1pm resolution is available. Laser power available is 0 to 10dBm). Assuming we want at least 100 points measured between adjacent minima and we want to be sure we can sweep through at least 2 FSR's then we can target FSR's in the range from 1nm to 20nm. Targeting 5 devices

total we will vary the length at a fixed width of 0.5um to achieve an imbalance in the MZI that will give rise to an FSR. We note that we could vary the width, which can modulate the effective index of the arms to create an imbalance in the MZI. But that is out of scope of the current study and will be saved as future work (Note, the meaning of a group index extracted from two arms with different effective indices is not clear to the author, but play sometimes leads to learning).

To estimate the difference in arm length required to achieve FSR's in the range of 1 to 20 nm we need to have a sense of the group index of the waveguide. Figures 1 and 2 show simulations of the electric field in a cross section of a silicon waveguide within silicon dioxide cladding. The geometry for the silicon is 220nm thick by 500nm wide. The simulations suggest that we should expect a group index of 4.195 for this set of geometries and materials. However manufacturing variability can lead to variation in both the waveguide height and width. Assuming the silicon thickness can vary between 215.3 and 223.1nm and the width can vary between 470 and 510 nm, we can expect the group index to range between 4.176 and 4.256. While we target our designs around the the 220nm by 500nm target geometries, we can expect FSR's in the range shown in the table 1.

		Min.	Max.	tar.	tar.	Max.	Min.
Device	dL (um)	ng	FSR (nm)	ng	FSR (nm)	ng	FSR (nm)
MZI20	30	4.176	19.2	4.195	19.1	4.256	18.8
MZI15	40	4.176	14.4	4.195	14.3	4.256	14.1
MZI6	100	4.176	5.8	4.195	5.7	4.256	5.6
MZI3	200	4.176	2.9	4.195	2.9	4.256	2.8
MZI1	585	4.176	1	4.195	1	4.256	1

Table 1: Design of experiments for imbalanced MZI's showing the anticipated range of FSR achieved at 1550nm.

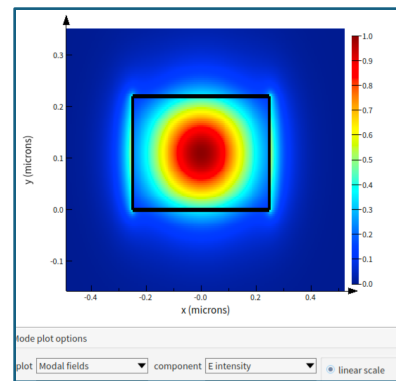


Fig 1: Simulated cross section of the electric field intensity in a 0.5um wide, 220nm thick waveguide slab.

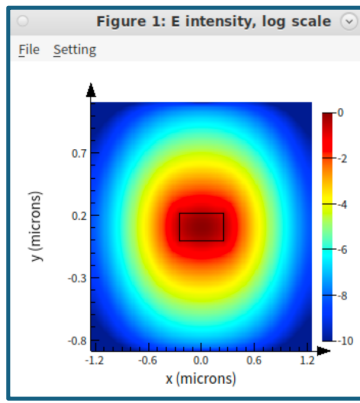


Fig 2: Logarithmic simulated cross section of the electric field intensity in a 0.5μm wide, 220nm thick waveguide slab showing the field is reasonably contained within the simulated region.

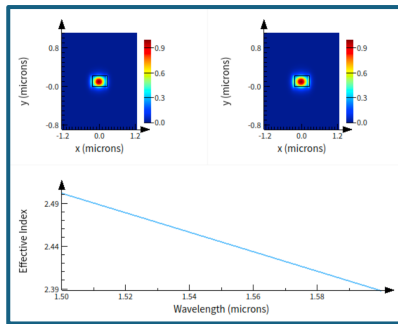


Fig. 3: Frequency analysis of above slab showing effective index as a function of frequency

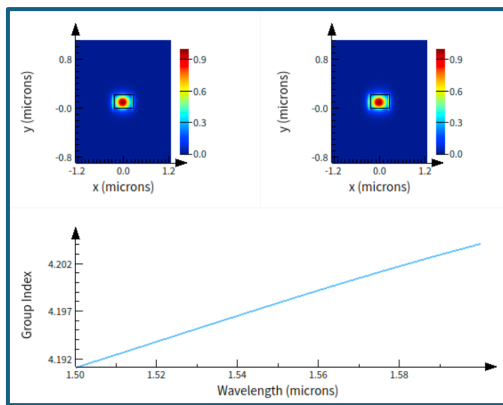


Fig. 4: Frequency analysis of above slab showing group index as a function of frequency

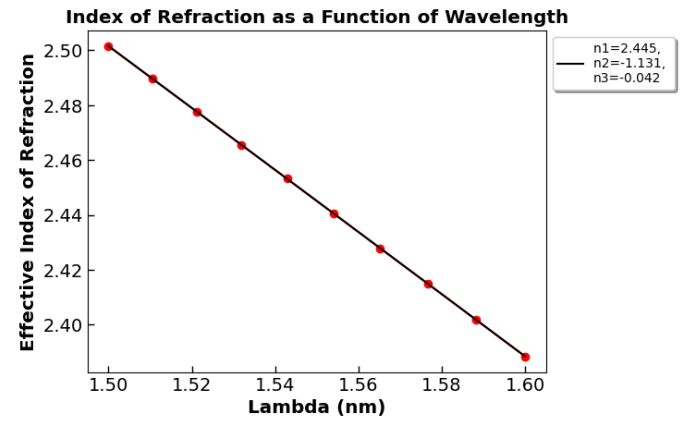


Fig 5: Compact model fit of simulated effective index of refraction as a function of wavelength. $n_{eff} = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2$ where n_1, n_2 and n_3 are shown in the plot legend for the fit.

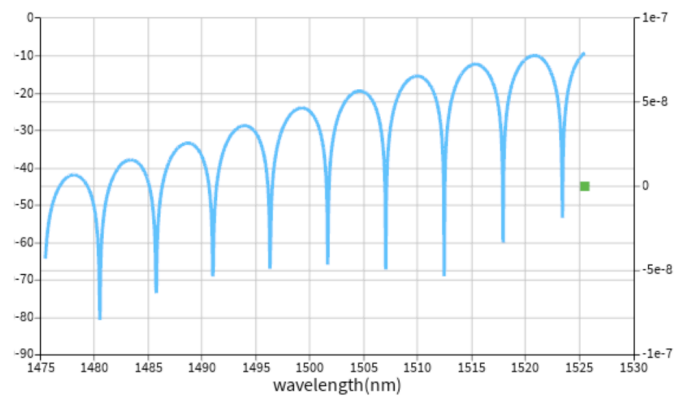


Fig. 6: Gain versus wavelength for the below MZI schematic using EBeam pdk elements. The path length difference in this case is 100nm and the simulation shows that we should expect an FSR of approximately 6nm.

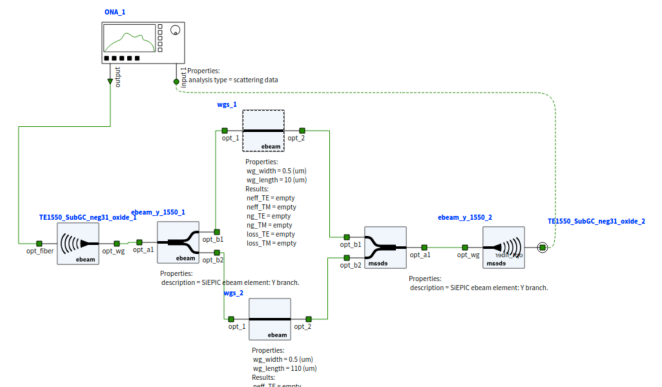


Fig. 7: Schematic view of an MZI used to simulate the above gain versus wavelength.

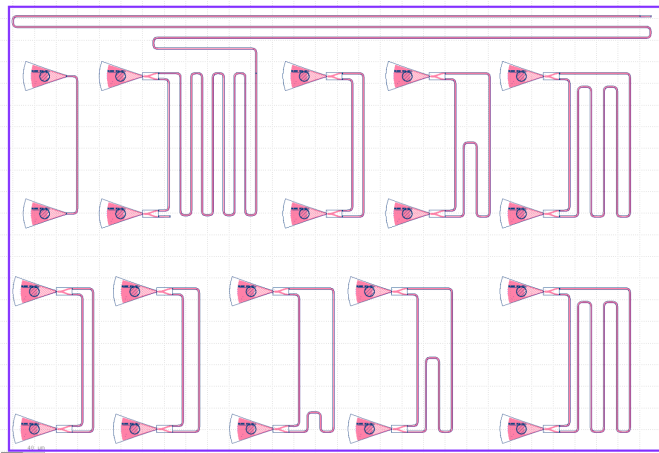


Fig. 8: Layout of the asymmetric MZI designs that will be used to extract group index values. The device on the upper left will be used as the primary de-embedding structure to remove loss due to the grating couplers. We note that a de-embedding structure consisting of a *balanced* MZI would also include losses due to the splitters. Future work may incorporate such a design.

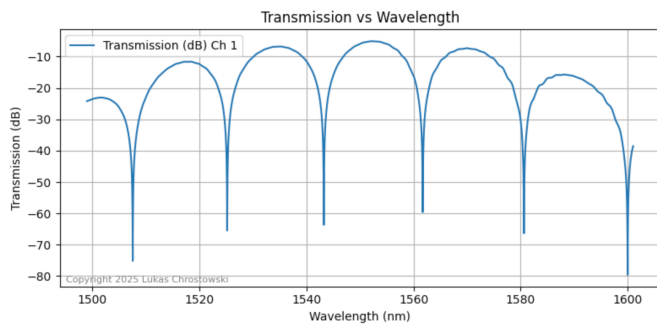


Fig. 9: Simulated output of MZI20 which is expected to have a 20nm FSR.

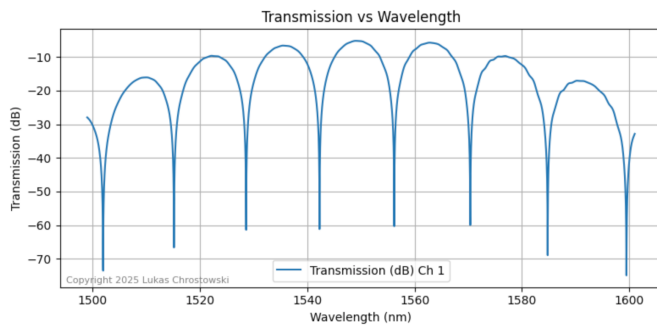


Fig. 10: Simulated output of MZI15 which is expected to have a 15nm FSR.

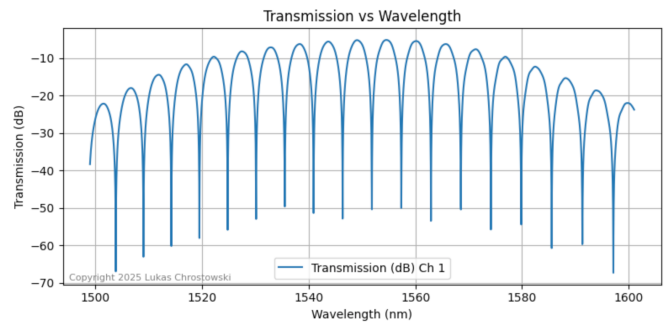


Fig. 11: Simulated output of MZI6 which is expected to have a 6nm FSR.

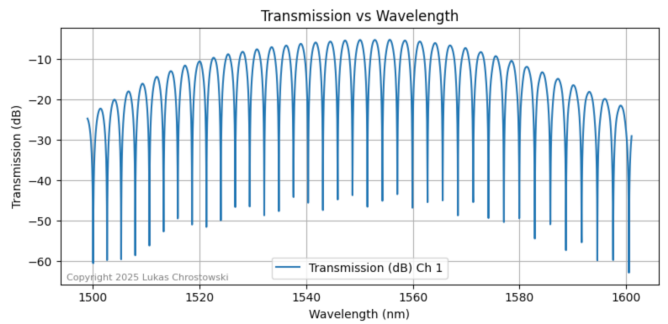


Fig. 12: Simulated output of MZI3 which is expected to have a 3nm FSR.

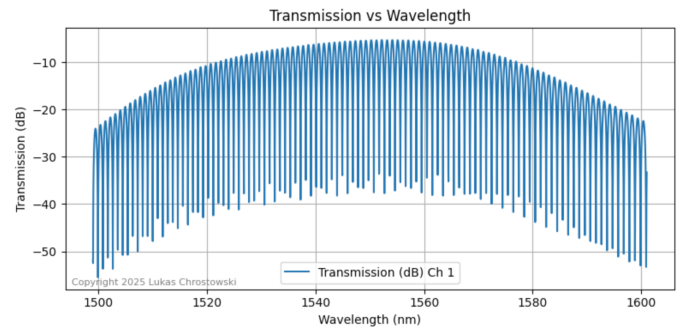


Fig. 13: Simulated output of MZI1 which is expected to have a 1nm FSR.

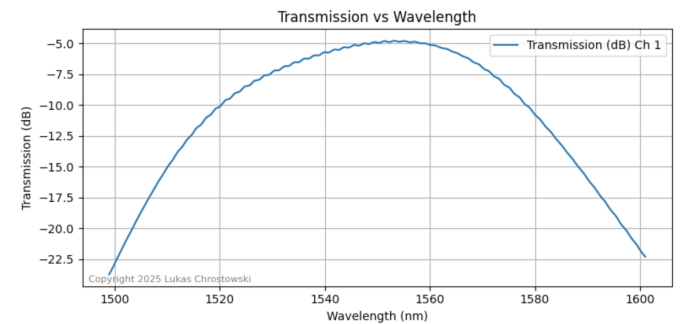


Fig. 14: Simulated output of the de-embedding structure. Note the loss increases away from the target wavelength of 1550nm.

Experimental Results

Figure 15 shows the transmission spectrum through the de-embedding structure (depicted in the upper left of figure 8). Note the peak transmission is around 1530nm, 20nm less than the anticipated peak of 1550nm. The yellow line in the figure is a fourth order polynomial fit to the data ranging from peak transmission to 10dB less than that transmission peak. Subsequent analysis is confined to this wavelength range.

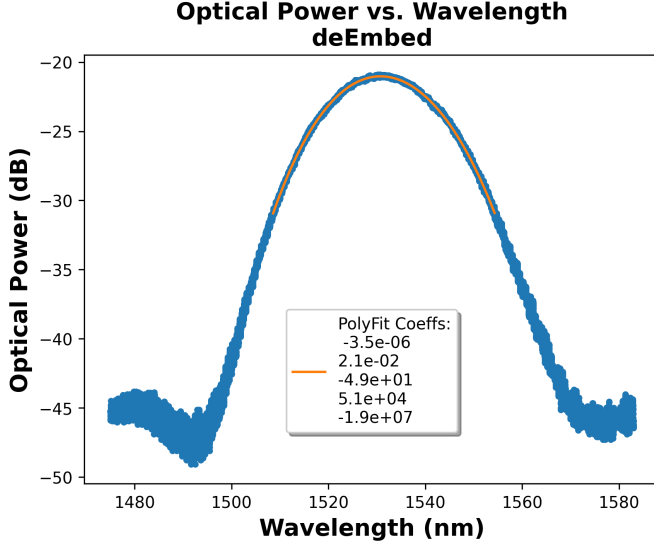


Fig. 15: Optical power transmitted through the de-embedding structure (upper left of figure 8). Note the peak transmission is around 1530nm, 20nm less than the anticipated peak of 1550nm. The yellow line in a fourth order polynomial fit to the data ranging from peak transmission to 10dB less than that transmission peak. Subsequent analysis is confined to this wavelength range.

Figure 16 shows the raw transmission spectrum of the device MZI6. We note that the peak transmission power is near 1530nm much like in the transmission spectrum of the de-embedding structure. We attribute the loss on either side of 1530nm to grating coupler losses. Confining our interests to the effects of the imbalanced MZI, rather than grating coupler losses, we subtract the polynomial fit of the de-embedding structure from the MZI transmission. To avoid measurement noise on the analysis we confine our analysis to the wavelength range that has less than 10dB loss relative to the peak transmission in the de-embedding structure. The result for MZI6 is shown in figure 17.

In figure 17 the magenta dots are data points (adjusted for grating coupler loss). The black line is a fit to the MZI transfer function.

$$F = 10 \log_{10} \left(\frac{1}{4} \left| 1 + \exp \left[-i \frac{2\pi n_{\text{eff}}}{\lambda} \Delta L - \frac{\alpha \Delta L}{2} \right] \right|^2 \right) + b$$

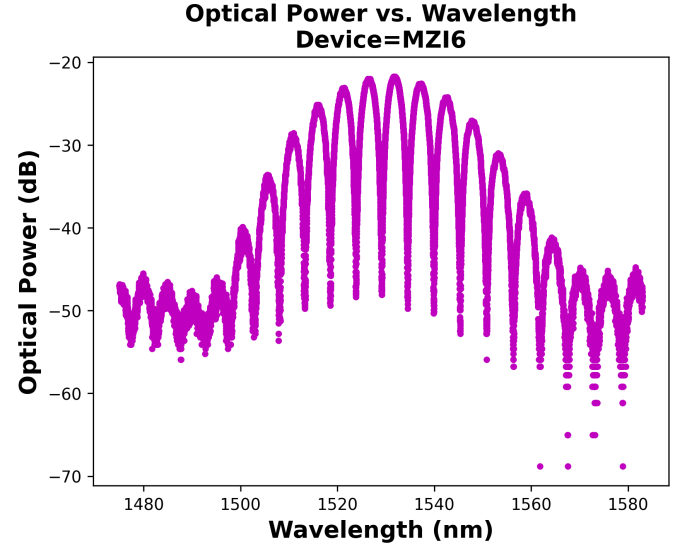


Fig. 16: Raw optical transmission spectrum of the device MZI6.

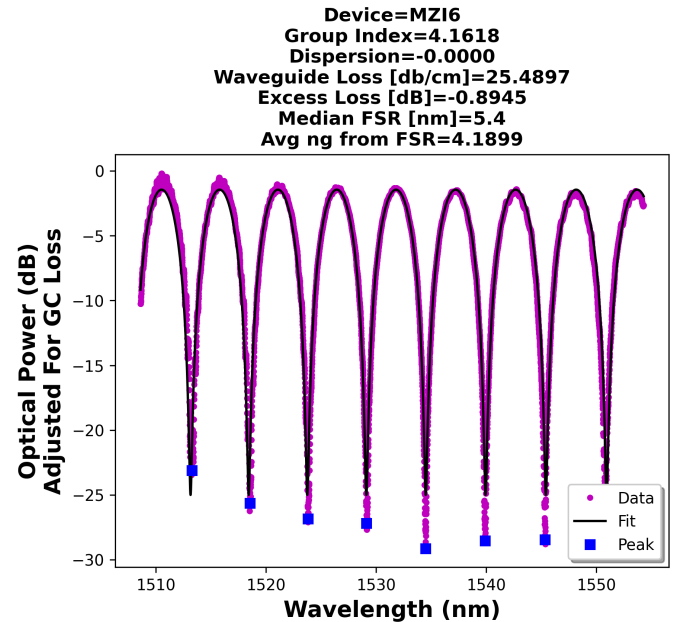


Fig 17: Optical transmission power as a function of wavelength for the device MZI6. The magenta dots are data points (adjusted for grating coupler loss). The black line is a fit to the MZI transfer function. The blue squares are peaks of destructive interference used to estimate initial fitting parameters for the MZI transfer function fit.

To estimate initial parameters for the MZI transmission spectrum, the destructive interference peaks were found using the function `find_peaks` from the Python `scipy.signal` library. The blue squares represent these peaks which allowed us to calculate an initial guess of the group index based on the FSR extracted from the distance between blue

peaks.

Using the initial parameters estimated by the distance between peaks we were able to fit the MZI transfer function for eight different MZI's encompassing five differences in arm length. Three of the path length differences were duplicated such that they had two nominally identical devices. Note that to successfully fit all devices we assumed the dispersion was equal to zero, which we believe is a reasonable assumption for the transverse electric (TE) optical modes explored in this study. From the transfer function fit we can estimate the waveguides group index, loss and excess loss. Results are shown for each path length difference in figures 17 through 20. The titles of each plot include the group index extracted from the MZI transfer function, the waveguide loss, excess loss, median FSR and the group index extracted from the FSR.

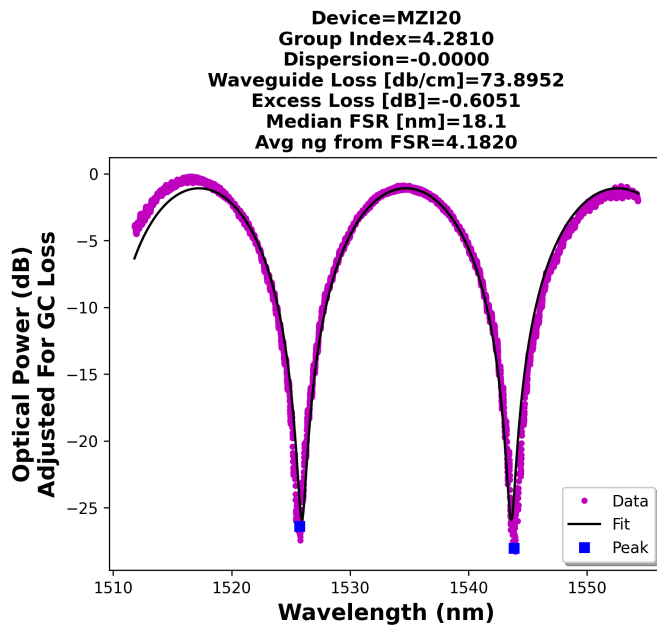


Fig. 17: Experimental data and fit of device MZI20.

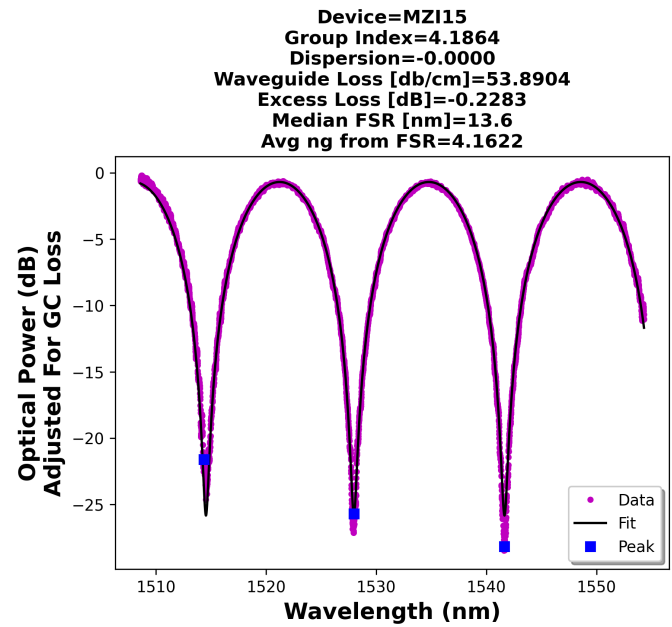


Fig. 18: Experimental data and fit of device MZI15.

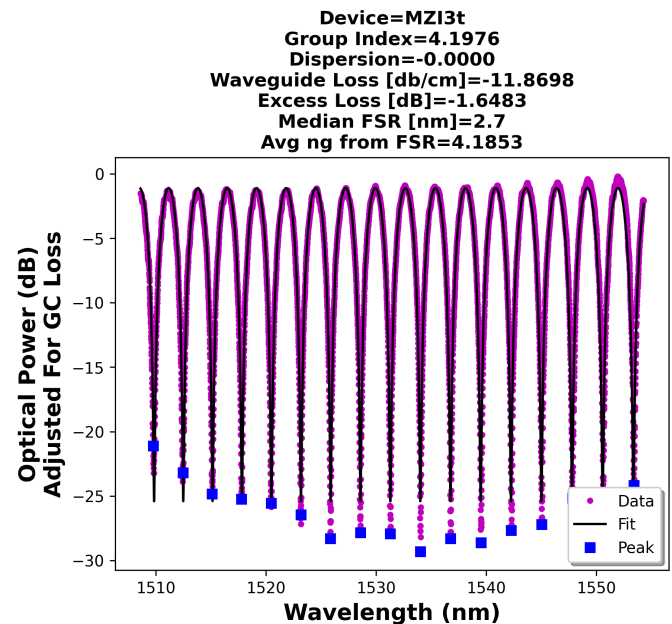


Fig. 19: Experimental data and fit of device MZI3.

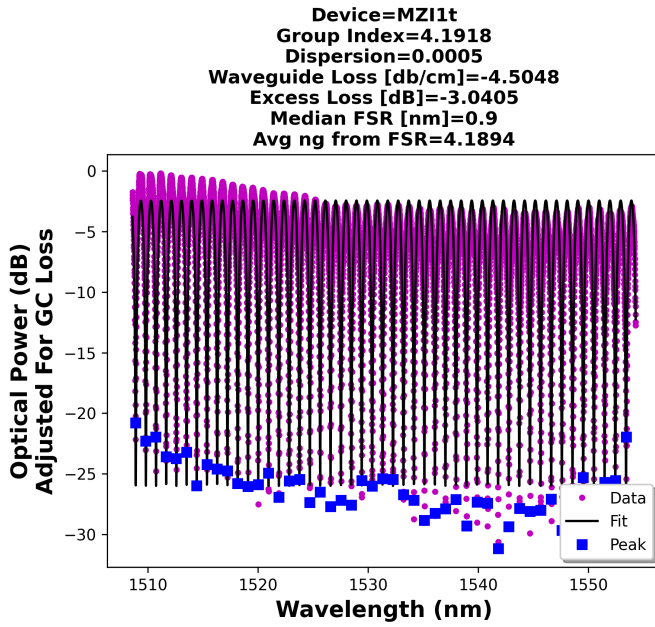


Fig. 20: Experimental data and fit of device MZI1t.

The measured data, adjusted for grating coupler loss, is shown in magenta. The peaks identified are shown by blue squares and the fit to the MZI transfer function is shown by a black line. The group index is extracted both from the average FSR and from the fit to the transfer function. These results are shown as a function path length difference (“DeltaL”) in figure 21. The green triangles represent the group index extracted from the average peak separation, or FSR. The red circles represent the group index extracted from the transfer function fit. The horizontal dashed lines represent the range of group indices that we would expect to extract given process variations as previously described. Note there are two nominally identical devices plotted at each of the path length differences 30, 200 and 585 μm respectively. Five out of eight devices fall within the expected range of group indices. The three devices that fall outside of the expected range deviate by less than 2%.

There is reasonably good agreement between the group index extracted from the distance between peaks (green triangles) and the transfer function fit (red circles). On nominally identical devices the results are very similar for path length differences of 200 and 585 μm . The device to device variation is the most pronounced for the smallest path length difference, 30 μm , of this device set. A speculative explanation for this is that process variations are more pronounced for the smaller path length difference compared to the larger path length differences where process variations can average out more easily.

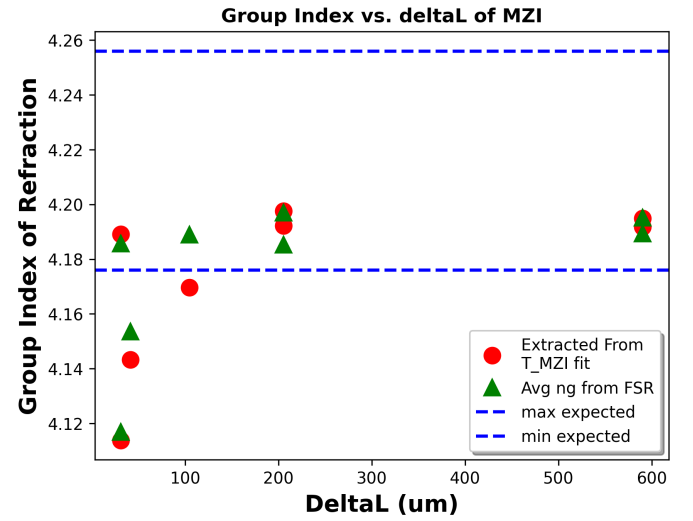


Fig 21: Group index of refraction as a function of difference in length of the two arms of a Mach Zehnder Interferometer for eight different devices. The red circles are extracted from fitting the optical transmission spectra of structures to their transfer function. The green triangles are average values calculated based on the FSR’s within the spectra. The dotted horizontal lines are from Lumerical Mode simulations and represent the range of group indices that we expect given typical manufacturing variations. Most of the devices fall within the expected range. Those that fall outside of the range, fall outside by less than 2%.

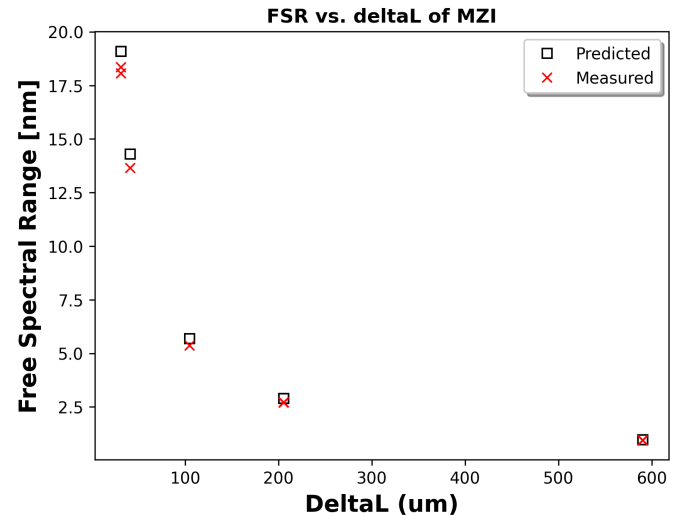


Fig. 22: Free spectral range (FSR) versus difference in MZI arm length. The black squares represent values predicted from theory and simulation. The red x’s represent the average of the measured values based on dip to dip distance in the optical spectra.

The predicted and measured free spectral ranges are plotted in figure 22 as a function of path length difference. In

general, agreement between the predicted and measured values is very good. At this scale one cannot even see the device to device differences in measured FSR for the long path length difference device (0.948 versus 0.942nm for $\Delta L=590\mu\text{m}$).

Conclusion

The group index of refraction was extracted from a set of asymmetrical Mach Zehnder Interferometers consisting of a silicon waveguide in silicon dioxide cladding. Simulations of the electric field across the waveguide cross section allowed us to estimate the group index. This enabled us to design a set of devices within an accessible manufacturing capability and experimental window such that we could measure the group index of real hardware. The designs were fabricated and measured. The subsequent analysis showed good agreement between simulation and experiment thus demonstrating a robust platform on which one can engineer a desired optical transmission spectrum. We found that longer path length differences fell within the expected range of group indices more often and had more repeatable results than short path length differences. We note that while the larger path length differences exhibited more predictable behavior, they come with a tradeoff of larger device area.

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Appendix 1 – Process Details (Plagiarized):

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (<http://www.appliednt.com/nanosoi>; Edmonton, Canada) which is based on direct-write 100 keV electron beam

lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 μm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 $\text{H}_2\text{SO}_4:\text{H}_2\text{O}_2$) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 μm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300°C. Reflectometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

Appendix 2– Measurement Details (Plagiarized):

To characterize the devices, a custom-built automated test setup [2, 6] with automated control software written in Python was used [3]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [4]. A 90° rotation was used to inject light into the TM grating couplers [4]. A polarization maintaining fibre array was used to couple light in/out of the chip [5].

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[2] Lukas Chrostowski, Michael Hochberg, chapter 12 in "Silicon Photonics Design: From Devices to Systems", Cambridge University Press, 2015

[3] <http://siepic.ubc.ca/probestation>, using Python code developed by Michael Caverley.

[4] Yun Wang, Xu Wang, Jonas Flueckiger, Han Yun, Wei Shi, Richard Bojko, Nicolas A. F. Jaeger, Lukas Chrostowski, "Focusing sub-wavelength grating couplers with low back reflections for rapid prototyping of silicon photonic circuits", Optics Express Vol. 22, Issue 17, pp. 20652-20662 (2014) doi: 10.1364/OE.22.020652

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[6] <http://mapleleafphotonics.com>, Maple Leaf Photonics, Seattle WA, USA.